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Institutions and Economic Growth: An Historical Introduction

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Summary. — In this essay the interdependence of political and economic institutions is examined against premises in neoclassical theories of economies, which maintain that population and savings are the principal determinants of economic growth. That institutions matter in the development of economies is the difference in assumptions I examine. I look at the institutional frameworks that preceded adoption of the United States Constitution and the Northwest Ordinance in early America. I then trace the precedents of these American institutions to the evolution of common law in England, where interests unconcerned with the Crown coalesced to form competing political units. I then contrast England's institutional development with the comparatively centralized enforcement mechanisms of Spain to examine the hypothesis that institutional differences produce economic consequences.

The objective of this essay is two-fold: (1) to develop a theoretical framework which focuses on the historical obstacles to economic growth; and (2) to briefly apply this framework to explore the contrasting characteristics of institutional change in early modern Britain and Spain and the downstream implications for North and Latin America.[1] In the sections that follow, I explore in: (1) the issues; (2) the nature of institutions; (3) the sources of institutional change; (4) the initial historical conditions in England and Spain; (5) English development; (6) Spanish development; and finally in section (7) the consequences for the New World. Of necessity, the historical sections are little more than outlines, illustrating the framework developed in the first sections.

1. THE ISSUES

I begin with one of the most cited but still misunderstood essays of our time. It is 25 years since Ronald Coase published "The Problem of Social Cost" (1960), but the impact of that essay has still not really penetrated the economics profession, or the development literature. Coase pointed out that the neoclassical model, which has served as the basis of economic reasoning for most scholars in the Western world, holds only under the severely restrictive assumption of zero transaction costs; but that with positive transaction costs, institutions matter. There are no institutions in the neoclassical world; and indeed in such a world growth is not a problem, its rate being simply a function of the number of children people have and the rate of saving.[2]

Ever since Adam Smith, economists have recognized that gains from trade are the key to the wealth of nations. Specialization and division of labor have made possible the improved productivity that arises from technological change, better resource allocation, and specialized production, the key underlying features of modern economies. What economists have not realized until recently is that the exchange process is not costless. Economists still misunderstand key dilemmas of economies and ignore the costs involved in exchange, assuming (as the standard neoclassicists do) that exchange is costless or unproductive (i.e., the classical notion of unproductive labor), or contending that such costs exist but are passive and therefore not important, or are neutral with respect to their consequences for economies.

In fact, the costs of transacting are the key to the performance of economies. There have always been gains from trade, as classical international trade theory has taught, but so too have there been obstacles to realizing these gains. If transport costs were the only obstacle, then we would observe through history an inverse rela-

*I wish to thank Elisabeth Case for improving this essay.

tionship between transport costs, on the one hand, and trade and exchange and the well-being of societies on the other. But recall that as early as the Roman Empire of the first two centuries AD trade was possible over a vast area, even with the transport costs of the time; and that after the end of the Roman Empire trade declined and probably the well-being of societies and individual groups declined as well. It was not that transport costs had risen; but that the costs of transacting had risen as regions expanded, and unified political systems that could effectively enforce rules and laws disappeared.

From the evidence of history, let us turn to the economies of the world today and observe the enormous disparity between the rich countries of the Western world and the poor countries of the Third World. It is not transport costs but the costs of transacting that are the key obstacles that prevent economies and societies from realizing well-being. We can understand why when we examine analytically the costs of transacting in different situations.

We begin with a simple model of personal exchange. In personal exchange individuals either engage in repeat dealings with others or otherwise have a great deal of personal knowledge about the attributes, characteristics, and features of each other. The measured transaction costs of a society where there is a dense social network of interaction are very low. Cheating, shirking, opportunism, all features that underlie modern industrial organization theory, are limited or indeed absent, because they simply do not pay. Under such conditions, norms of behavior are seldom written down. Formal contracting does not exist, and there are few formal specific rules. However, while measured transaction costs in such societies are low (although unmeasured costs of societal cooperation in tribal societies may indeed be high), production costs are high, because specialization and division of labor are limited to the extent of markets that can be defined by personal exchange.

At the other extreme from personal exchange is a world of specialized interdependence in which the well-being of individuals depends upon a complex structure characterized by individual specialization and hence by exchange ties that extend both in time and space. A pure model of this world of impersonal exchange is one in which goods and services or the performance of agents is characterized by many valued attributes, in which exchange takes place over time, and in which there are not repeat dealings. Under these forms of exchange, the costs of transacting can be high, because there are problems both in measuring the attributes of what is being exchanged and problems of enforcing the terms of exchange; in consequence there are gains to be realized by engaging in cheating, shirking, opportunism, etc.[3] In order to prevent such activity, elaborate institutional structures must be devised that constrain the participants and so minimize the costly aspects I have just described. As a result, in modern Western societies we have devised formal contracts, bonding of participants, guarantees, brand names, elaborate monitoring systems, and effective enforcement mechanisms. In short, we have well-specified and well-enforced property rights. The result of all this is that resources devoted to transacting (although small per transaction) are large, while the productivity associated with the gains from trade is even greater; and high rates of growth and development have characterized Western societies. Of course these institutions depend on a much more complex institutional structure that makes possible the specification and enforcement of property rights, which in turn allow transactions to occur and productivity gains from modern technology to be realized.

Increasing specialization and division of labor necessitate the development of institutional structures that permit individuals to take actions that involve complex relationships with other individuals both in terms of personal knowledge and over time. The evolution of more complex social frameworks will not occur if such institutional structures cannot reduce the uncertainties associated with such situations. So, institutional reliability is essential, because it means that even as the network of interdependence caused by the growth of specialization widens we can have confidence in outcomes that are necessarily increasingly remote from our personal knowledge.[4]

The institutional requirements that are necessary in order to be able to realize the productivity gains associated with the model of impersonal exchange outlined above entail both the development of efficient products and factor markets and of a medium of exchange with reliable features. The establishment of such a set of property rights will then allow individuals in highly complex interdependent situations to be able to have confidence in their dealings with individuals of whom they have no personal knowledge and with whom they have no reciprocal and ongoing exchange relationships. This is only possible as the result, first, of the development of a third party to exchanges, namely government, which specifies property rights and enforces contracts; and second of the existence of norms of behavior to constrain the parties in interaction, which will permit exchange where high measurement costs,

even with third party enforcement, pose problems with respect to opportunism, cheating, etc.

But why isn't it automatic to develop more and more complex institutions that will enable us to handle more complex interdependence? Indeed, much of the literature of game theory and stories of institutional development imply that the progress of primitive societies to the status of modern Western societies should be automatic and unilinear. The answer is quite clear. The breakdown of personal exchange is not just the breakdown of a dense communication network, but is the breakdown of communities of common ideologies and of a common set of rules in which all believe. The rise of impersonal rules and contracts means the rise of the state, and with it unequal distribution of coercive power. This provides the opportunity for individuals with superior coercive power to enforce the rules to their advantage, regardless of their effects on efficiency. That is, rules will be devised and enforced on behalf of the interests of the politically advantaged but they will not necessarily lower the costs of transacting in total.

In fact, one of the most evident lessons from history is that political systems have an inherent tendency to produce inefficient property rights which result in stagnation or decline. There are two basic reasons for this result. First, the revenue that can be raised by rulers may be greater with an inefficient structure of property rights that can, however, be effectively monitored, and therefore taxed, than with an efficient structure of property rights with high monitoring and collection costs.[5] Second, rulers can seldom afford efficient property rights, since such rights can offend many of their constituents and hence jeopardize the security of others' rights. That is, even when rulers wish to promulgate rules on the basis of their efficiency consequences, survival will dictate a different course of action, because efficient rules can offend powerful interest groups in the polity.[6]

2. THE NATURE OF INSTITUTIONS

Institutions are rules, enforcement characteristics of rules, and norms of behavior that structure repeated human interaction. Hence, they limit and define the choice set of neoclassical theory. We are interested not in the institutions *per se*, but in their consequences for the choices individuals actually make.

Constitutions, statute and common laws, and contracts specify in formal terms the rules of the game, from the most general constitutional ones to the specific terms of a particular exchange. Rules (and their enforcement) are constrained by the costliness of measuring the characteristics or attributes of what constitutes rule compliance or violation. Hence, the technology of measurement of all the dimensions (sight, sound, taste, etc.) of the human senses has played a critical role in our ability to define property rights and other types of rules. Moreover, since we receive utility from the various attributes of goods and services rather than from the entities themselves, it is the costliness of measuring the separable dimensions that is critical in this study. The relationship between the benefits derived from rule specification and the costs of measurement not only has been critical in the history of property rights (common property vs. private property) but is at the heart of many of the issues related to the structure and effectiveness of enforcement.

If it were costless to measure the performance of agents or the attributes of goods and services as well as the terms of exchange, then enforcement would not be a problem. We would be back in the neoclassical world of the instantaneous exchange of a unidimensional good or service. But because measurement is costly and the parties to exchange stand to gain by receiving the benefits without incurring all of the costs of exchange, not only is enforcement typically imperfect, but the structure of the enforcement process will affect outcomes and hence choices. Let me elaborate both points.

Enforcement is typically imperfect for two reasons: measurement is costly; and the interests of principals and agents are not identical. The costliness of measurement implies that at the margin the benefits from additional monitoring or policing will be balanced against the incremental costs. Moreover, as I shall discuss below, the marginal benefits and costs of policing will be weighed against those of investing at the margin in ideological persuasion. Rules are enforced by agents (police, foremen, judges, juries, etc.), and therefore the standard problems of agency theory obtain. It is important to stress here that both the structure of the enforcement mechanism and the degree of imperfection of enforcement are important in the choices that are made.[7]

Rules and their (imperfect) enforcement are not the complete story. If they were, the modeling of institutions and hence the costs of transacting could be made, at this stage of our knowledge, much more precise. But norms of behavior also matter; and we know very little about them.

As a first approximation, norms are informal constraints on behavior that are in part derivative of formal rules; that is they are extensions of such

rules and apply to specific issues. These informal procedures, deriving as they do from formal organizational structures and agendas, are important but still relatively easy to analyze.[8] Much more important, norms are codes of conduct, taboos, standards of behavior, that are in part derived from perceptions that all individuals form both to explain and to evaluate the world around them. Some of these perceptions are shaped and molded by organized ideologies (religions, social and political values, etc.). Others are honed by experience, which leads to the reaffirmation or rejection of earlier norms.

However they are formed, and however they evolve, norms play a critical role in constraining the choice set at a moment of time and in the evolution of institutions through time. They are important at a moment of time precisely because of the costliness of measurement and the imperfect enforcement of rules. To the degree that individuals believe in the rules, contracts, property rights, etc., of a society, they will be willing to forego opportunities to cheat, steal or engage in opportunistic behavior. In short, they live up to the terms of contracts. Conversely, to the degree that individuals do not believe in the rules, regard them as unjust, or simply live up to the standard wealth maximizing behavioral assumption we typically employ in neoclassical economics, the costs of contracting, that is transaction costs, will also increase. Empirical evidence suggests the price we are willing to pay for our convictions is a negatively sloped function, so that ideological attitudes are less important as the price increases; but both the slope of the function and shifts in the functions are subjects about which we know very little.[9]

The foregoing paragraphs suggest that ideas and values matter at a moment of time. They do so because of "slack in the system," "agency costs," "consumption on the job," etc., all of which result from the costliness of measurement and enforcement. But how do they change through time? Certainly fundamental changes in relative prices lead not only to rule (and enforcement) changes; but to changes in ideas and values, and the rate of these two kinds of change may be markedly different. This subject will be explored below, but first let me raise some specific issues about institutions, transaction costs, and the consequent choices of the "players," which bear on the subject of this essay.

Let me start with a quotation from Bill Riker.

> . . . Every time I convince myself that I have found an instance in which constitutional forms do make a difference for liberty, my discovery comes apart in my hands. . . . Professor Ostrom believes that at least part of the reason we believe we are a free people is that we have certain constitutional forms, but it may just as easily be the case that the reason we have these constitutional forms is that we are a free people.[10]

Now let me quote Bill Riker, a decade later.

> The Constitution was in a formal sense a necessary condition for the achievement, that is had the Articles [of Confederation] survived, the nation would not have flourished. To see this, note the Constitution was, in a formal sense, necessary for political unity and the consequent political dominance of the United States, first in America and its expansion westward, then in the western hemisphere by restraining European empirical expansion and finally in the world, helping to destroy in two world wars Western European monarchies and empires and later in countering the Soviet empire. All this depended on political unity. Yet without the Constitution, North America might very well have been as Balkanized as South America.[11]

Rules themselves are not a sufficient condition for determining outcomes even though they are, on occasion, critical. Indeed, the second quotation is from an extraordinarily insightful analysis of the formation of the Constitution, in which Riker makes the convincing case that its creation results from a rather unique concatenation of events that, as the quote implies, changed the destiny of the 13 confederated states. But it is important to remember that a number of Latin American countries patterned their constitutions after that of the United States with radically different results.

It may be a slight exaggeration to assert that enforcement is always imperfect, but this statement focuses our attention on a critical and neglected aspect of economic history, which is the essential role that third party enforcement of contracts has played in human economic progress. There is a large literature in the new industrial organization on self-enforcing contracts, etc.; but as with so much of modern economics, it misses the larger issues involved in exchange in a specialized world. Personal exchange solves the problems of contract fulfillment by repeat dealings and a dense network of social interaction. But the key to the high-income societies of the Western world is still the one that Adam Smith propounded more than 200 years ago. Increasing specialization and division of labor necessitate the development of institutional structures that permit individuals to take actions involving complex relationships with other individuals far removed from any personal knowledge and extending over long periods of time. This is only possible with a third party to exchange, government, which specifies property rights and enforces contracts.

Let me emphasize that while third party enforcement is far from perfect, there are vast differences in the relative certainty and effectiveness of contract enforcement, temporally over the past five centuries in the Western world, and more currently between modern Western and Third World countries. The evolution of government from its medieval Mafia-like character to that embodying modern legal institutions and instruments is a major part of the history of freedom. It is a part that tends to be obscured or ignored because of the myopic vision of many economists, who persist in modeling government as nothing more than a gigantic form of theft and income redistribution.

In a recent paper, Robert Axelrod tells the story of Alexander Hamilton writing, on the last night of his life, all of the reasons why he should not engage in a duel with Aaron Burr.[12] They were rational and overwhelmingly convincing reasons but in the end not sufficient to overcome the dishonor that he perceived would result from a code of conduct that required such a solution of disputes among gentlemen. Axelrod's purpose in telling this story was to illustrate that norms of behavior, which are not legal rules, are enforced by the attitudes and behavior of others in the society.

But this is surely only part of that complex of ideas, customs, dogmas, values, ethical standards, etc., which make up our understanding of the world around us, establish our normative standards, and help define the choices we make. While some norms are externally enforced, others are internally enforced codes of conduct, like honesty or integrity. It would be an immense contribution to have a testable general theory of the sociology of knowledge and therefore an understanding of the way overall ideologies emerge and evolve.[13] In the absence of such a theory, we can still derive an important and potentially testable implication about norms at a more specific microlevel of analysis, which is derived from an understanding of institutions. Specifically, the structure of rules and their enforcement help define the costs we bear for ideologically determined choices; the lower the costs, the more will ideas and ideologies matter. Let me provide three illustrations.

A basic paradox of public choice is that individual votes do not matter, but lots of people still vote. Brennan and Buchanan point out this dilemma in a recent paper but do not satisfactorily resolve it.[14] Surely one of the things voters are doing is expressing strongly held convictions at low cost to themselves. Moreover, in the aggregate votes do of course matter. The expression "putting your money where your mouth is" characterizes neither voters nor for that matter academics. Both can afford to be in the literal sense, *irresponsible*.

Recently a large volume of literature has grown up around agency theory applied to legislators, with the voter as principal and the legislator as agent. Empirical work suggests that legislators frequently vote their own conviction rather than principals' interests.[15] Other empirical work suggests that the institutional structure of Congress permits legislators to engage in strategic voting behavior that effectively conceals their true objectives.[16]

Finally, judges with lifetime tenure can and do vote their own convictions, as even the most casual study of courts in general and the Supreme Court in particular testifies. Moreover, this is not an accident. The constitutional provisions, as interpreted by the Marshall Court (1801–35), were deliberately designed to remove judges from interest group pressures.[17]

However possible it is to show that ideas matter, it is much more difficult to trace the way they have evolved. For example, the demise of slavery is simply not explicable in an interest group model. Surely the micro argument described above is important to understanding its end. That is, most of those who voted for its elimination, either directly or indirectly, paid few or no costs; they could simply express their abhorrence of one human being owning another. There was no institutional way for the slave owner to buy off the voters. On the other hand, the *way* in which the anti-slavery movement grew (and frequently was used by interest groups) so that it could lead to these votes is a much more complex story.

It is time we took stock. The neoclassical model describes the output of an economy as a function of the quantity and costs of a set of inputs of land, labor, capital and entrepreneurship, given some production function derived from the state of technological knowledge. But this formulation is, if not incorrect, largely misleading, since if that were all there were to the output of societies, they would all be rich (given again some rather noncontroversial behavioral assumptions). Restated more usefully, the costs of production are a function of the costs of the traditional inputs listed above and of the costs of transacting.

It is important to stress that economic growth can and has occurred as a consequence of increasing productivity. That is, either technological change or changes in institutions (political or economic) that affect the specification and enforcement of property rights may increase productivity.

The measurement of transaction costs is beset with all of the problems of measurement in traditional national income accounting. To the extent that transactions occur in the market economy, they can be measured.[18] However, costs of transacting arising from queuing, waiting, rationing, bribery, etc., which are substantial in all economies but particularly in Third World and socialist economies, are unmeasured.

In considering historical measurement we must remember that the costs of transacting have frequently been so high that no production or exchange occurred. The lack of institutions and instruments to facilitate production and exchange in factor and product markets (as well as the existence of institutions designed to raise the costs of transacting) resulted in predominantly personalized (and localized) production and trade. The development of institutions to facilitate transacting is marked not only by expansion of production and exchange in particular factor markets, but by a subsequent decline in transaction costs as the institutions develop. The dramatic decline in real interest rates in 17th century Netherlands and early 18th century Britain followed the development of capital market institutions; it is probably the best quantitative measure (and the most critical indicator) of improving productivity in the transactions sector.

The foregoing discussion profiles transaction costs in a growing economy. The emergence of political institutions that specify "efficient" property rights and provide increasingly effective enforcement should show up in terms of the development of economic institutions to facilitate market exchanges. As a result, costs per transaction will be falling; but the size of the transactions sector in the aggregate will be a growing proportion of GNP, as increasing specialization and division of labor multiplies the aggregate volume of exchanges. This is precisely the pattern for the United States, where the measured size of the transaction sector in 1870 is about one-quarter of GNP and in 1970 is almost one-half.

3. THE SOURCES OF INSTITUTIONAL CHANGE

There are two issues I wish to address on institutional change: what causes the change; and what determines its path? In neither case have I a completely satisfactory answer.

Before we can turn to these two issues, we must examine the role institutions play in reducing uncertainty in human interaction, since it is this stabilizing role of institutions which separates clearly the framework of analysis being developed here from the traditional neoclassical approach. We can most readily understand the difference if we have ever visited foreign countries and attempted to "do business" with them. We will find that of necessity we must learn their "way of doing things." The structural forms of human interaction that characterize societies are a combination of rules, enforcement features, and norms of behavior. Until we learn what these are, the costs of transacting are high. Once we understand them, we can effectively communicate and engage in varieties of social, political, and economic exchange. The function of institutions is to provide certainty in human interaction, and this is accomplished by the inherent features of rules and norms. Rules are typically nested in a hierarchical structure, each more costly to change. But even in the absence of the hierarchical institutional structure, the status quo typically has an advantage over changes in a variety of political structures, as a consequence of agenda control and committee structure.

It is norms of behavior, however, that probably provide the most important sources of stability in human interaction. They are extensions, elaborations, and qualifications of rules that have tenacious survival ability, because they become an integral part of habitual behavior. The reduction of uncertainty, in consequence, makes possible regular human interaction; but it in no way implies that the institutions are efficient, only that they dampen the consequences of relative price changes.

But institutions do change, and fundamental changes in relative prices do lead to institutional change. Historically, population change has been the single most important source of relative price changes, though technological change (including and importantly, changes in military technology) and changes in the costs of information have also been major sources. Moreover, as briefly noted in the previous section, changes in norms of behavior, while certainly influenced by relative price changes, are also influenced by the evolution of ideas and ideologies.

A stylized characterization of the process of institutional change could proceed as follows: as a result of a relative price change, one or both parties to an exchange (political or economic) perceives that he (they) could do better with an altered agreement (contract). Depending on his relative (and presumably changed) bargaining power, he will, as a consequence of the changed prices, renegotiate the contract. However, contracts are nested in a hierarchy of rules. If the renegotiation involves alteration of a more fundamental rule, he (or they) may find it

worthwhile to devote resources to changing the rule; or gradually, over time, the rule or custom may simply become ignored and/or unenforced. Agenda power, free-rider problems, and norms of behavior will add meat (and lots of complications) to this skeletal outline.

An important distinction in this argument is made between absolute bargaining power and changes at the margin. To illustrate this distinction, I turn to the medieval world. The "agreement" between lord and serf on the medieval manor reflected the overwhelming power of the lord vis-à-vis the serf. But changes at the margin, as a consequence of 14th century population decline, altered the opportunity costs, increased the relative bargaining power of serfs, and led to the gradual evolution of copyhold.[19]

While institutional evolution may proceed in the above manner, without explicit intent or design, dramatic changes in the rules (or their enforcement) occur as well. The gathering in Philadelphia in 1787 is a clear example. Riker, in the essay referred to above, makes clear that the instigators of the convention were Federalists and that their opposition both misunderstood and misjudged their ability to write and ratify a new constitution. Indeed, it was the promised addition of the Bill of Rights as the first order of business under the new constitution that made ratification possible (Riker, forthcoming). Perhaps it is worth noting that the writers were "gentlemen" and that their promise was both believed and carried out.

A special note should be made of the role of military technology in institutional change. Not only have changes in military technology resulted in different, efficient (survival) sizes of political units, but, as in the story that follows, they have consequently induced fundamental changes in other institutions, so that fiscal revenues essential to survival could be realized.

The second issue of institutional change is what determines the direction of change. From what must have been quite common origins several million years ago or even as recently as the hunting and gathering societies that predate the "agricultural revolution" in the 8th millenium BC, we have evolved in radically different directions (and at radically different rates). How have we evolved such divergent patterns of social, political, and economic organization? To consider a specific example, as I will do in the subsequent sections of this paper, how do we explain the divergent paths of British and Spanish development, both at home and in the contrasting histories of North and South America?

I believe the answer lies in the way that institutional structures evolve. The closest (although by no means perfect) analogy is the way we perceive that the common law evolved. It is precedent-based law: past decisions become embedded in the structure of rules, which marginally change as cases arise evolving some new or, at least in the terms of past cases, unforeseen issue, which when decided becomes, in turn, a part of the legal framework. However, I don't intend to imply by this analogy that the result is "efficient." In fact, as we shall see, Spanish institutional evolution moved in the direction of stagnation.

Let me illustrate evolution by reference to a specific act, which was almost as important as the Constitution in United States history. This was the Northwest Ordinance, passed by the Congress (when it was under the Articles of Confederation) in 1787, at the very time that the Constitutional Convention was meeting in Philadelphia. It was the third act dealing with the issues that arose with respect to the settlement, governance, and integration of the vast lands to the west in the new nation. Where did the rules incorporated in these acts come from and how were they arrived at?

The Ordinance is quite simple and brief. It provided for rules of inheritance and fee-simple ownership of land; it set up the basic structure of the territorial governments and provided for the mechanisms by which territories gradually became self-governing. Additionally, it made provisions for when a territory could be admitted as a state. Then there was a series of Articles of Compact, in effect a bill of rights for the territories (i.e., provisions for religious freedom, the writ of *habeas corpus*, trial by jury, bailment, enforcement of contract and compensation for property). There were additional provisions about good faith to the Indians, free navigation on the Mississippi and the St. Lawrence, public debt, land disposal, the number of states that could be divided up within the Northwest Territory, and finally a provision prohibiting slavery (though the return of runaway slaves was provided for) in the territories.

It is easy to trace most of the provisions. Inheritance laws, fee-simple ownership of land, and many of the provisions of the Bill of Rights had evolved and become a part of the rules of political units of the colonies during the previous 150 years (described in more detail below). However, some that were precedent-based in the territorial charters became controversial as new issues arose. The size of new states and the conditions admitting them were two such issues. Precedents originated in the provisions of charters and the Articles of Confederation.

Controversies arose from the implications of shifting political powers under the new government of the Constitution. The prohibition of slavery in new states, for example, appears to be the result of vote-trading between the authors of the Northwest Ordinance and the writers of the Constitution; slavery would be prohibited in the Northwest territories if counting slaves as three-fifths of a person were accepted in the Constitution, a policy that would increase the representation of Southern slave states in Congress (already a major issue of the period).[20]

The Northwest Ordinance provided the basic framework dictating the pattern of expansion of the American nation over the next century. While its provisions were at times modified by new issues and controversies, it provided a clear, "path-dependent" pattern of institutional evolution. It is only understandable in terms of precedent, new issues, and the bargaining strength of the parties. It is essential to note that precedents not only defined and determined many of the provisions but also dictated the existing agenda, decision rules, and method of resolution. The larger point of this illustration is that we can only understand historical change by modeling the way institutions evolved through time. That brings us to the following brief outline of English and Spanish institutional change, from the 1500s to the 19th century in North America and Latin America.

4. INITIAL HISTORICAL CONDITIONS IN ENGLAND AND SPAIN

Despite the similarities between England and Spain (discussed below) at the beginning of the 16th century, the two countries had evolved very differently. Spain had just emerged from seven centuries of Moorish domination of the peninsula. It was not really a unified country. Although the marriage of Ferdinand and Isabella brought Castile and Aragon together, they continued to maintain separate rules, Cortes (parliaments), and policies. England, in contrast, had developed a relatively centralized feudalism as a result of the Norman conquest and, with the Battle of Bosworth (1485), had recently established the power of the Tudors.

Yet, in common with the rest of the emerging European nation states, they each faced a problem with far-reaching consequences: that a ruler required additional revenue to survive. The tradition was that a king was supposed to live on his own, which meant that the income from his estates, together with the traditional feudal dues, were his total revenue. The changes in military technology associated with the effective use of the cross-bow, long-bow, pike, and gunpowder enormously increased the cost of warfare and led to a fiscal crisis first described by Joseph Schumpeter (1919). In order to get more revenue, the king had somehow to make a bargain with constituents. In both countries, this initially led to the development of some form of representation on the part of the constituents in return for revenue, and in both countries, the wool trade became a major source of crown revenue. Thereafter, the stories diverge. We can better appreciate this divergence in the framework of a very simple model of the state, consistent with the framework developed in the previous sections of this essay.[21]

The king acts like a discriminating monopolist, offering to different groups of constituents "protection and justice," or at least the reduction of internal disorder and the protection of property rights, in return for tax revenue. Since different constituent groups have different opportunity costs and bargaining power with the ruler, there result different bargains. But economies of scale also exist in the provision of these (semi) public goods of law and enforcement. Hence, total revenue is increased, but the division of the incremental gains between ruler and constituents depends on their relative bargaining power; changes at the margin in either the *violence* potential of the ruler or the opportunity costs of the constituents will result in re-divisions of revenue increments. Moreover, the rulers' gross and net revenues differ significantly as a result of the necessity of developing agents (a bureaucracy) to monitor, meter, and collect the revenue; and all the inherent consequences of agency theory obtain here. The initial institutional structure that emerged in order to solve the fiscal crisis therefore looked similar in all the emerging nation states of Europe. A representative body (or bodies) of constituents, designed to facilitate exchanges between the two parties, was created. To the ruler it meant the development of a hierarchical structure of agents, which was a major transformation from the simple (if extensive) management of the king's household and estates to a bureaucracy monitoring the wealth and/or income of the king's constituents. Let us see how this initial framework evolved in the two cases.

5. ENGLISH DEVELOPMENT

The tension between rulers and constituents (although that would hardly describe the situation at Runnymede in 1215) surfaces with the

Magna Carta; but the fiscal crises come to a head with Edward I and Edward III during the Hundred Years War. Stubbs in his *The Constitutional History of England* (1896) summarizes the consequences.

> The admission of the right of parliament to legislate, to inquire into abuses, and to share in the guidance of national policy, was practically purchased by the monies granted to Edward I and Edward III.

A logical consequence was that in the 16th century under the Tudors the structure of Tudor government was revolutionized, as Geoffrey Elton has described in *The Tudor Revolution in Government* (1953). This revolution transformed the government from an elaborate household structure into a bureaucracy increasingly concerned with overseeing and regulating the economy. It had early on been the wool trade which had served as the basis for a good deal of tax revenue; and, as told by Eileen Powers (1941), the wool trade involved a three-way relationship between the exporters, the wool growers as represented in Parliament, and the Crown. In this agreement, the Merchants of the Staple achieved a monopoly of the export trade as well as a depot in Calais. Parliament received the right to set the tax and the Crown obtained the revenue. In England the combined mix of the growth of the wool trade, the development of fee-simple ownership in land, and the development of arable lands and new crops imported from the Dutch contributed to an expansion of agriculture. At the same time, in the nonagricultural sector the economy became increasingly diversified. Although the Tudors continued to attempt to control the economy and to freeze the structure of economic activity into guilds and monopolistic activities, their efforts were relatively ineffective. They were ineffective because (1) the statutes only covered existing industries, so that new industries escaped rule; (2) despite opposition by town guilds, industries moved to the countryside and effectively escaped guild control; (3) the control of wages and laborers in the Statute of Artificers of 1563 was only partially and sporadically enforced; and (4) enforcement in the countryside was typically in the hands of unpaid justices of the peace who had little incentive to enforce the law.

The cloth trade therefore grew in the countryside. The interplay between the expansion of diverse economic activities escaping from guild restrictions and the pressures for the development of parliamentary control over the sovereign came to a head with the Stuarts, with the fumbling efforts of James I, the continuing fiscal crises that occurred under Charles I, and the articulate opposition of Coke and others. It was Coke who insisted that the common law was the supreme law of the land, and who repeatedly incurred the anger of James I. It was Coke who led the parliamentary opposition in the 1620s, which established common law control over commercial law. By the end of Elizabeth's reign, a changing benefit-cost pattern of economic activity was emerging with the widening of domestic and foreign markets; the result was the expansion of voluntary organizations in the form of joint stock companies, and growing resentment against the crown-sponsored monopolies which excluded private companies from many of these growing markets. *Darcy vs. Allein* was only the most celebrated case reflecting this ongoing struggle to create a set of rights that would be outside the control of the monarchy. Passing the Statute of Monopolies was just another step in the ongoing process.

Yet the issue of the supremacy of Parliament hung in the balance for much of the 17th century. As the struggle continued, Parliament not only attempted to wrest from the King's control the granting of monopolies (as in the Statute of Monopolies), but also to protect itself from the King's wrath by establishing religious, civil, and political freedoms as well (such as the Petition of Right in 1628). It distorts the story, however, to think of it as a clear-struggle between an absolutist "oriented" king and a unified Parliament concerned with economic, civil, and political liberties.[22] As the Civil War attests, a complex of religious, economic, and political interests coalesced into armed camps. Moreover, the winning coalition one day could be in the minority the next day. Hence, there was persistent interest and concern with broadly based and impersonally guarded rights.

The final triumph of Parliament was produced in 1689, and in rapid consequence came a set of economic institutions reflecting the relatively increasing security of property rights. The creation of the Bank of England (1694) and the development of new financial instruments led to a dramatic decline in the cost of transacting and have been described as the English financial revolution. Both institutions and consequent falling transaction costs reflect increased security of the time dimension of property rights, a dimension critical to both a long-term capital market and to economic growth itself.

In terms of the very simple political model outlined in Section 2, the original trade of certain rights to Parliament in return for revenues was a product of the fiscal crises of the Hundred Years War. In the 17th century, the Tudors' tripartite

arrangement between the Crown, Parliament, and merchants granted further rights to Parliament in return for tax revenues (and still further rights were granted to the Commons by Henry VIII for support in his controversial seizure of church properties). In consequence, the Tudors required an organized bureaucracy to oversee tax collection and to regulate other parts of the economy. The triumph of Parliament in 1689 simply shifted the locus of decision making to Parliament, which raises the issue of why Parliament would not then proceed to act just like the King. Ekelund and Tollison argue:

> Higher costs due to uncertainty and growing private returns reduced industry demands for regulation and control in England. All this strengthened the emergent Constitutional democracy, which created conditions making rent-seeking activity on the part of both monarch and merchant more costly. When the locus of power to rent-seeking shifted from the monarch to Parliament . . . the costs of supply and regulation through legislative enactment rose, for reasons suggested by the theory of public choice.[23]

The framework of institutional evolution I have described suggests a somewhat more complicated story than Ekelund and Tollison provide. They assert, "While it is flattering to think that intellectuals affect public policy — surely they do to some extent — it seems completely out of character for economists to think that intellectual arguments could affect real magnitudes so strongly."[24] But the embedding of economic and political freedoms in the law, the interests of principals (merchants, etc.) in greater degrees of freedom, and the ideological considerations that swept England in the 17th century combined to play a role in institutional change.

6. SPANISH DEVELOPMENT

While the major steps in Spanish institutional evolution are not in question, nor is the final result, I do not believe that the specific steps along the way have been as clearly delineated as in the English story. (It should be emphasized that I am not nearly as familiar with the Spanish literature as with the English; but it is my impression that an explicit analysis of the evolution of property rights and their political origins has not been a focus of Spanish economic history.) However, some sketch is possible.

Prior to the union of Ferdinand and Isabella, the kingdom of Aragon (comprising approximately Valencia, Aragon, and Catalonia) had a very different character than Castile. Aragon had been reconquered from the Arabs in the last half of the 13th century and had become a major commercial empire extending into Sardinia, Sicily, and part of Greece. The Cortes, reflecting the interests of merchants "had already secured the power to legislate and even to limit the king's power to issue legislation under certain conditions" (Veliz, 1980, p. 34). In contrast, Castile was continually engaged in warfare, either against the Moors or in internal strife. While the Cortes existed, it was seldom summoned,

> and as nobility and clergy were exempt from financial exactions that could conceivably join them with representatives of the town in resisting additional levies by the Crown, they did not pose (that is the towns did not pose) a credible challenge . . ." (Veliz, 1980, p. 35).

In the 15 years after their union, Isabella succeeded in gaining control not only over the unruly warlike barons but over church policy in Castile as well. The result was a centralized monarchy in Castile; and it was Castile that defined the institutional evolution of both Spain and Latin America.

A major source of fiscal revenues was the Mesta (the sheep-herders guild), which in return for the right to migrate with their sheep across Castile provided the Crown with a secure source of revenue, but also with consequences adverse to the development of arable agriculture and the security of property rights, as well as with soil erosion.[25]

Within Castile the other chief source of revenue was the *alcaba*, a sales tax. But as the Spanish empire grew to become the greatest empire since Roman times, its major sources of revenue were increasingly external, derived from Sicily, Naples, the low countries, and the New World. Control internally over the economy and externally over the far-flung empire entailed a large and elaborate hierarchy of bureaucrats armed with an immense out-pouring of royal edicts. Over 400,000 decrees had been issued concerning the governance and economy of the Indies by 1635, an average of 2,500 a year since Columbus sailed first to the Indies (Veliz, 1980, p. 43). Designed to provide minute regulation of the economy, guilds also provided a vehicle for internal economic regulation. Price ceilings were imposed on grain and state-owned trading companies, and monopolistic grants provided control of external trade.

As the military costs of controlling the empire outstripped revenues (which declined with the revolt of the Netherlands and the gradual decrease in receipts of treasure), the Crown raised the internal tax (*alcaba*) from 1.2% to 10% and repeatedly went into bankruptcy, which is resolved through the seizure of properties and financial assets. The consequence was the de-

cline of the Spanish economy and economic stagnation.[26]

In terms of the foregoing model of the polity, the bargaining position of the Crown, vis-à-vis the Cortes, shifted in favor of the Crown and consequently resulted in the decline of the Cortes. The governance structure then became a large and elaborate bureaucracy and there were endless efforts by the Crown to control its far-flung agents. Indeed, the history of the control of the Indies is an elaborate story in agency theory, beginning as early as Isabella's recision of Columbus' policies toward the Indians in 1502. Distance magnified the immense problem of monitoring agents in the New World; but despite the dissipation of rent at every level of the hierarchical structure, the Crown maintained effective control over the polity and over the economy of the New World.[27]

7. CONSEQUENCES FOR THE NEW WORLD

It is likewise much easier to trace the institutional evolution of the English North American colonies than their Latin American counterpart. The initial conditions are in striking contrast. English America was formed in the very century when the struggle between Parliament and the Crown was coming to a head. Religious diversity, as well as political diversity in the mother country, was paralleled in the colonies. In the Spanish Indies, conquest came at the precise time that the influence of the Castilian Cortes was declining. The conquerers imposed a uniform religion and a uniform bureaucratic administration on the existing agricultural society.[28]

In the English colonies there was substantial diversity in the political structure of crown proprietary and charter colonies. But the general development in the direction of local political control and the growth of assemblies was clear and unambiguous. Similarly, the Navigation Acts placed the colonies within the framework of overall British imperial policy, and within that broad framework the colonists were free to develop the economy. Indeed, the colonists themselves frequently imposed more restrictions on property rights than did the mother country. (The exception was the effort of proprietors to obtain quit-rents from settlers in proprietary colonies, such as that of Lord Penn. The problem of enforcement and collection in the context of the availability of land resulted in very indifferent success.)

In the Spanish Indies, a bureaucracy detailed political and economic policy.

> . . . the *residencia* was the principal means employed by the king to keep viceroys and other functionaries under control. On the expiration of their term of office, all officials had to undergo the official investigation of their conduct. The fear of the residencia was frequently an incentive to serve the monarch well; it also limited any autonomous inclination of ambitious civil servants in the periphery of the empire (Veliz, 1980, p. 73).

As for the economy, the Marquis of Pombal, who was the Secretary of State for Foreign Affairs and War, and who "ruled like a virtual dictator" Portugal and its empire from 1755–77, is said to have stated:

> I find it absolutely necessary to bring all the commerce of this kingdom and its colonies into companies, then all merchants will be obliged to enter them, or else desist from trading, for they certainly may be assured that I know their interests better than they do themselves and the interest of the whole kingdom (Veliz, 1980, pp. 108–109).

Some merchants and the Lisbon Chamber of Commerce protested.

> Pombal properly dissolved the chamber and had several of its leading members imprisoned; the rest were regrouped under direct government supervision into a *Juntado Commercio* that dutifully approved all the minister's decisions (Veliz, 1980, p. 109).

The French and Indian War (1755–63) is the familiar breaking point in American history. British efforts to impose (very modest) taxes on colonial subjects, as well as to curb westward migration, produced a violent reaction that led through a sequence of steps to the Revolution, the Declaration of Independence, the Articles of Confederation, the Northwest Ordinance, and the Constitution: a sequence of institutional expressions that formed a consistent evolutionary institutional pattern, despite the precariousness of the process.

In the Spanish Indies the recurrent crises were over the efficiency and control of the bureaucratic machinery. The decline under the Hapsburgs and the revival efforts under the Bourbons led to restructuring of the bureaucracy and even some liberalization of trade (under the Bourbons) within the empire. But the control of agents was a persistent problem, compounded by efforts of the Creoles to take over the bureaucracy in order to pursue their own interests. To whatever degree the wars of independence in Latin America were a struggle between colonial control (of the bureaucracy and consequent polity and economy) and imperial control, the struggle was imbued with ideological overtones that stemmed for the American and French

revolutions. Independence brought United States inspired constitutions, but with radically different consequences. In contrast to the United States, in Latin America, "The imaginative federal schemes and courageous attempts at decentralization had one thing in common after the first few years of republican independence: they were all tried but none worked. Some were disastrous; none survived."[29]

The contrasting histories of North and South America are perhaps the best comparative case that we have of the consequences of divergent institutional paths for political and economic performance. We are only just beginning to extend economic and political theory to the study of institutions.[30] I hope this historical introduction gives some indication of the promise of this approach for the study of economic history and economic growth.

NOTES

1. This essay is a combination of two earlier papers of mine, "Institutions, transaction costs and economic growth," prepared for a World Bank conference on the Political Economy of Poverty, Equity and Growth, and subsequently published in *Economic Inquiry* (North, 1988a), and "Institutions, economic growth and freedom: An historical introduction," prepared for a Liberty Fund Symposium on Economic, Political and Civil Freedom, and published in *Freedom, Democracy and Economic Welfare*, edited by Michael Walker (North, 1988b).

2. See North (1981), Chapter I, for an elaboration of the implications of the noeclassical model for economic growth.

3. See Lancaster (1966), Becker (1965), Cheung (1974), North (1981), and Barzel (1982) for an elaboration of the theory underlying this argument.

4. Heiner (1983) explores these issues in an interesting essay.

5. The granting of monopolies to guilds in Colbert's France produced this result.

6. For an elaboration of these points, see "A Neoclassical Theory of the State," in North (1981) and North (1984).

7. Oliver Williamson's (1975) approach differs from the approach developed in this essay for several reasons, but principally because he takes imperfect enforcement as a given (otherwise opportunism would never pay) rather than recognize that both the characteristics of the enforcement process and the degree of imperfection are essential in modeling institutions and to the costs of transacting.

8. See for example Shepsle and Weingast (1986).

9. See Kalt and Zupan (1984) for discussion of these issues.

10. Riker (1976).

11. Riker (forthcoming).

12. Axelrod (1985).

13. The immense literature on the subject from Marx and Mannheim to Merton is not very convincing, although Robert Merton's chapters written in 1949 are still a good summary of the state of the art.

14. Brennan and Buchanan (1983).

15. Kalt and Zupan (1984).

16. Denzau, Riker and Shepsle (1985).

17. Landes and Posner (1975) provide an interest group model of the Supreme Court, but the evidence simply does not support such an argument. See Buchanan (1975) and North (1978).

18. See North and Wallis (1987) for a lengthy discussion of the issues and measurement of the transaction sector in the American economy between 1870 and 1970.

19. See North and Thomas (1973) for a description of this process.

20. For an elaboration of these issues, see North and Rutten (1987).

21. This simple "neoclassical theory of the state" is elaborated in North (1981), Chapter 3.

22. Moreoever, it should be noted that the rights that Parliament had in mind were those of the nobility and the gentry.

23. Ekelund and Tollison (1981), p. 149.

24. Ekelund and Tollison (1981), p. 151.

25. The history of the Mesta (Klein, 1920) is an exception to my assertion that the history of Spanish property rights has not been told. For a summary of the effects of the Mesta on the development of property rights in Spain, see North and Thomas (1973), Chapter 10.

26. For a more detailed account and sources, see North and Thomas (1973), Chapter 10.

27. For a more detailed account, see Veliz (1980), Chapter 3: "The Regalist Indies."

28. "The Indian population was subdued and subordinated by the new ruling class of Encomenderos. But the Encomendero class itself was the target of a royal program that reduced its political significance by installing a class of state office holders" (Lang, 1975, p. 220).

29. Veliz (1980), p. 151. Veliz provides a country-by-country summary account of the decline of democratic government and the revival of the "centralist" bureaucratic structure and tradition in Latin America (Chapter 7).

30. See North (1986) for an analysis of the "New Institutional Economics."

REFERENCES

Axelrod, Robert, "An evolutionary approach to norms," *American Political Science Review*, Vol. 80, No. 4 (1986), pp. 1095–1111.

Barzel, Yoram, "Measurement and the organization of markets," *Journal of Law and Economics*, Vol. 25 (1982), pp. 27–48.

Becker, Gary, "A theory of the allocation of time," *Economic Journal*, Vol. 65 (1965), pp. 493–517.

Brennan, Geoffrey, and James Buchanan, "Voter choice — Evaluating political alternatives," *American Behavioral Scientist*, Vol. 28 (1983), pp. 185–206.

Buchanan, James, "Comment on: 'The independent judiciary in an interest group perspective'," *Journal of Law and Economics*, Vol. 18 (1975), pp. 903–906.

Cheung, Stephen, "A theory of price control," *Journal of Law and Economics*, Vol. 17 (1974), pp. 53–69.

Coase, Ronald, "The problem of social cost," *Journal of Law and Economics*, Vol. 3 (1960), pp. 1–44.

Denzau, Arthur, William Riker, and Kenneth Shepsle, "Farquharson and Fenno: Sophisticated voting and homestyle," *American Political Science Review*, Vol. 79, No. 4 (1985), pp. 1117–1134.

Elton, Geoffrey, *The Tudor Revolution in Goverment*, (Cambridge: Cambridge University Press, 1953).

Ekelund, Robert, and Robert Tollison, *Mercantilism as a Rent Seeking Society* (College Station, TX: Texas A & M Press, 1981).

Heiner, Ronald, "The origins of predictable behavior," *American Economic Review*, Vol. 73 (1983), pp. 560–595.

Kalt, Joseph, and Mark Zupan, "Capture and ideology in the economic theory of politics," *American Economic Review*, Vol. 74, No. 3 (1984), pp. 279–300.

Klein, Julius, *The Mesta* (Cambridge, MA: Harvard University Press, 1920).

Lancaster, Kelvin, "A new approach to consumer theory," *Journal of Political Economy*, Vol. 74 (1966), pp. 132–157.

Landes, William, and Richard Posner, "The independent judiciary in an interest group perspective," *Journal of Law and Economics*, Vol. 18 (1975), pp. 875–902.

Lang, James, *Conquest and Commerce: Spain and England in America* (New York: Academic Press, 1975).

Merton, Robert, *Social Theory and Social Structure* (Glencoe, IL: Free Press, 1949).

North, Douglass, "Institutions, transaction costs and economic growth," *Economic Inquiry*, Vol. 25 (July 1988a).

North, Douglass, "Institutions, economic growth and freedom: An historical introduction," in Michael Walker (Ed.), *Freedom, Democracy and Economic Welfare* (Vancouver, BC: The Fraser Institute, 1988b).

North, Douglass, "The New Institutional Economics," *Journal of Institutional and Theoretical Economics*, Vol. 142 (1986), pp. 230–237.

North, Douglass, "Government and the cost of exchange in history," *Journal of Economic History*, Vol. 44 (1984), pp. 255–264.

North, Douglass, *Structure and Change in Economic History* (New York: W. W. Norton, 1981).

North, Douglass, "Structure and performance: The task of economic history," *Journal of Economic Literature*, Vol. 16 (1978), pp. 963–978.

North, Douglass, "Sources of productivity change in ocean shipping, 1600–1850," *Journal of Political Economy*, Vol. 76 (1968), pp. 953–970.

North, Douglass, and Andrew Rutten, "The Northwest Ordinance in historical perspective," in David Klingaman and Richard Vedder (Eds.), *Essays in the Economy of the Old Northwest* (Athens, OH: The Ohio University Press, 1987).

North, Douglass, and Robert Thomas, *The Rise of the Western World* (Cambridge: Cambridge University Press, 1973).

North, Douglass, and John Wallis, "Measuring the transaction sector in the American economy, 1870–1970," in Stanley L. Engerman and Robert E. Gallman (Eds.), *Long-Term Factors in American Economic Growth*, Vol. 51 of the Income and Wealth Series (Chicago: University of Chicago Press, 1987).

Powers, Eileen, *The Wool Trade in English Medieval History* (London: The Clarendon Press, 1941).

Riker, William, "The lessons of 1787," *Carnegie Papers on Political Economy*, Vol. 7 (forthcoming).

Riker, William, "Comment on: 'American experiment in constitutional choice'," *Public Choice*, Vol. 27 (1976), pp. 13–15.

Schumpeter, Joseph, "Crisis of the tax state" (1919), reprinted in *International Economic Papers* (1934), pp. 5–8.

Shepsle, Kenneth, and Barry Weingast, "The institutional foundation of committee power," Political Economy Working Paper #105 (St. Louis, MO: Washington University, 1986).

Stubbs, William, *The Constitutional History of England* (London: The Clarendon Press, 1896).

Veliz, Claudio, *The Centralist Tradition of Latin America* (Princeton, NJ: Princeton University Press, 1980).

Williamson, Oliver, *Markets and Hierarchies* (New York: Free Press, 1975).